Brazilian E-Commerce Sales Analysis

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BUSINESS PROBLEM & OBJECTIVES

This Project analyzes Brazilian e-commerce sales data using SQL. The dataset contains customer information, orders, products, payments, reviews, and seller. This project helps understand: Sales Performance, Customer behavior, Top-selling products, Payment analysis, Delivery performance and seller analysis.



Business Questions:

- Revenue generated
- Sales trends
- Best products
- Top sellers
- Preferred payment methods
- Delivery efficiency

Dataset Overview

Dataset: Brazilian E-Commerce Public Dataset by Olist

To Create Database

```
Create Database e_commerce;
```

To Access Database

```
use e_commerce;
```

For Data I have taken From Kaggle

https://www.kaggle.com/datasets/olistbr/brazilian ecommerce

How to Import Data

- Right-click the table name in the left navigation sidebar.
- Select Table Data Import Wizard.
- Browse and select your JSON or CSV source file.
- Define whether to target a new table or an existing table.
- Verify column mappings and execute the import.

Database Schema & Relationships

Customers -> Orders -> Order Items -> Products/Sellers
Orders -> Payments

Tables Created: **Customers, Orders, Order Items, Payments, Products, Sellers.**

Table	Purpose
Customers	Customer details
Orders	Order information
Payments	Revenue
Order Items	Product sales
Products	Product details
Sellers	Seller details





Table Creation

Customers Table

```
create table customers (
customer_id varchar(100) primary key,
customer_unique_id varchar(100),
customer_zip_code_prefix varchar(20),
customer_city varchar(100),
customer_state varchar(20)
);
```

Orders Table

```
create table orders (
order_id varchar(100) primary key,
customer_id varchar(100),
order_status varchar(50),
order_purchase_timestamp datetime,
order_approved_at datetime,
order_delievered_carrier_date datetime,
order_delievered_customer_date datetime,
order_estimated_delivery_date datetime,
foreign key (customer_id) references customers(customer_id)
);
```

Order Payments Table

```
create table order_payments (
order_id varchar(100),
payment_sequential int,
payment_type varchar(50),
payment_installment int,
payment_value decimal(10,2)
);
```

Products Table

```
create table products (
product_id varchar(100) primary key,
product_category_name varchar(100),
product_name_length int,
product_description_length int,
product_photos_qty int,
product_weight_g int,
product_length_cm int,
product_height_cm int,
product_width_cm int
);
```

Sellers Table

```sql
create table sellers (
seller_id varchar(100) primary key,
seller_zip_code_prefix varchar(20),
seller_city varchar(100),
seller_state varchar(20)
);
```

Order Items Table

```sql
create table order_items (
Order_id varchar(100),
order_item_id int,
product_id varchar(100),
seller_id varchar(100),
shipping_limit_date datetime,
price decimal(10,2),
freight_value decimal(10,2),
foreign key (order_id) references orders(order_id),
foreign key (product_id) references products(product_id),
foreign key (seller_id) references sellers(seller_id)
);
```

SQL Concepts :

. Primary Key

. Joins

. Aggregate functions

. Where

. Order by

. Group by

Overall Business Performance

Find the Total Sales from Revenue Generated by Brazilian E-commerce

```
select round(sum(payment_value),2) as total_sales
from order_payments;

/* total sales      -----'16008872.12' */
```

→Reaching a revenue milestone of over 16 million indicates a significant volume of transactions, suggesting the platform has well-established user base.

Find the Count of Total Orders From Customers of Brazilian E-commerce

```
select count(*) as total_orders
from orders;

/* total orders ----96461 */
```

→This indicates a high level of customer trust and platform scalability

Insights:

- Total Revenue Generated
- Total Orders

The platform generated significant revenue from thousands of customer orders, indicating strong marketplace activity.

Top 10 Selling Product

```sql
select p.product_id, op.payment_value
from order_payments op
join order_items oi on op.order_id = oi.Order_id
join products p on oi.product_id = p.product_id
group by p.product_id, op.payment_value
order by op.payment_value desc
limit 10;
```

```
/* top 10 selling products
'5769ef0a239114ac3a854af00df129e4', '13664.08'
'19936fa4f614ee0590d3b77ac83fd648', '7274.88'
'489ae2aa008f021502940f251d4cce7f', '6929.31'
'c183fd5d2abf05873fa6e1014ed9e06c', '6081.54'
'a6492cc69376c469ab6f61d8f44de961', '4950.34'
'c3ed642d592594bb648ff4a04cee2747', '4764.34'
'259037a6a41845e455183f89c5035f18', '4681.78'
'a1beef8f3992dbd4cd8726796aa69c53', '4513.32'
'6cdf8fc1d741c76586d8b6b15e9eef30', '4175.26'
'dd113cb02b2af9c8e5787e8f1f0722f6', '4163.51' */
```

→ The Result show individual product is High-Value Transaction
→ This analysis identifies the Top 10 Selling products by individual transaction value. By joining payment and product datasets, we can pinpoint high-ticket items that contibute significantly to single-order revenue

Monthly Order Analysis

```sql
select year(order_purchase_timestamp) as year,
month(order_purchase_timestamp) as month,
count(order_id) as count_of_orders
from orders
group by year, month
order by year, month;
```

```
/* monthly orders analysis
'2016', '9', '1'
'2016', '10', '270'
'2016', '12', '1'
'2017', '1', '748'
```

→ Order volume increased steadily over time, indicating business growth

Payment Type Analysis

```sql
select payment_type, count(*) as total_transaction, round(sum(payment_value),2) as total_sales
from order_payments
group by payment_type
order by total_sales desc;
```

```
/* payment type analysis
'credit_card', '76795', '12542084.19'
'boleto', '19784', '2869361.27'
'voucher', '5775', '379436.87'
'debit_card', '1529', '217989.79'
'not_defined', '3', '0.00' */
```

Insights:

- Most preferred payment method
- Revenue contribution by payment type

Credit card transactions dominate customer purchases, showing preference for installment-based payments.

State wise Customer Count

```sql
select customer_state, count(*) as count_of_customer_id
from customers
group by customer_state
order by count_of_customer_id;
```

```
/*state-wise Customer count
'RR', '46'
'AP', '68'
'AC', '81'
'AM', '148'
'RO', '253'
```

→ There is a massive disparity between the top-performing states and the rest. This suggests business operations or marketing efforts are currently highly regionalized

Seller Performance

```sql
select seller_id, round(sum(price),2) as most_sellings
from order_items
group by seller_id
order by most_sellings desc
limit 10;
```

```
/*seller performence
'4869f7a5dfa277a7dca6462dcf3b52b2', '176198.19'
'53243585a1d6dc2643021fd1853d8905', '171752.69'
'fa1c13f2614d7b5c4749cbc52fecda94', '154671.02'
'4a3ca9315b744ce9f8e9374361493884', '153036.78'
'7c67e1448b00f6e969d365cea6b010ab', '148863.40'
'7e93a43ef30c4f03f38b393420bc753a', '135239.12'
'da8622b14eb17ae2831f4ac5b9dab84a', '126540.39'
'7a67c85e85bb2ce8582c35f2203ad736', '111599.07'
'1025f0e2d44d7041d6cf58b6550e0bfa', '109049.70'
'955fee9216a65b617aa5c0531780ce60', '100204.13' */
```

→ The query identifies the top 10 Sellers on the platform. And All top 10 sellers have surpassed the 100,000 revenue threshold. This suggests a healthy high tier segment within marketplace

Average Order

```sql
select round(avg(payment_value),2) as total_average
from order_payments;

/*Average Order value is '154.10' */
```

→ This value '154.10' serves as critical indicator of customer purchasing behavior, suggesting that customers are likely purchasing multiple items per transaction or higher-prices individual products.
→ Average Order Value directly impacts bottom-line profitability; monitoring this helps in evaluating the effectiveness of pricing strategies and promotional campaigns over time.

Delivery Delay Analysis

```sql
select order_id,
case
when datediff(order_delievered_customer_date, order_estimated_delivery_date)>0
then datediff(order_delievered_customer_date, order_estimated_delivery_date)
else 0
end as delivery_delays
from orders
where datediff(order_delievered_customer_date, order_estimated_delivery_date)>0;
```

```
/* the delivery delays
000e906b789b55f64edcb1f84030f90d    2
0017afd5076e074a48f1f1a4c7bac9c5    4
001c85b5f68d2be0cb0797afc9e8ce9a    8
001d8f0e34a38c37f7dba2a37d4eba8b    2
0030d783f979fbc5981e75613b057344    22
00324b3eda39ba5ecce3945823e3594c    3
0032d07457ae9c806c79368d7d9ce96b    12
00335b686d693c7d72deeb12f8e89227    32
```

→The CASE statement specifically filters for instances where the actual delivery occurred after the estimate (datediff > 0)
→ The Result Grid shows significant delays.
→ Such substantial discrepancies between estimated and actual delivery dates likely lead to high customer dissatisfaction.

Top 10 Cities by Orders

```sql
select c.customer_city, op.payment_value
from customers c
join orders o on c.customer_id = o.customer_id
join order_payments op on o.order_id = op.order_id
group by c.customer_city, op.payment_value
order by op.payment_value desc
limit 10;
```

```
/*Top 10 cities by orders
rio de janeiro  13664.08
vila velha  7274.88
campo grande    6929.31
vitoria 6922.21
marilia 6726.66
divinopolis 6081.54
araruama    4950.34
maua    4764.34
joao pessoa 4681.78
sao paulo   4513.32 */
```

→ The data suggests a strong performance in major Brazilian metropolitan areas. Resources for logistics and marketing should likely remain focused to maintain this momentum

Revenue by State wise

```sql
select c.customer_state, round(sum(op.payment_value),2) as total_revenue
from customers c
join orders o on c.customer_id = o.customer_id
join order_payments op on o.order_id = op.order_id
group by c.customer_state
order by total_revenue desc;
```

```
/* revenue by state
 'SP', '5768355.09'
'RJ', '2055878.41'
'MG', '1819029.99'
'RS', '861054.35'
'PR', '781919.55'
'SC', '595361.91'
'BA', '591270.60'
'DF', '346146.17'
'GO', '334294.22'
```

→ The highest revenue concentrations are in SP, RJ and MG. These three states represent the core of the business's financial intake
→The data suggests a highly centralized customer base or market presentation

How many orders delivered lately

```sql
select count(*) as late_deliveries
from orders
where order_delievered_customer_date > order_estimated_delivery_date;

/* late deliveries are '7826' */
```

→Late deliveries are a leading cause of low Customer Satisfaction scores and increased customer support tickets, these 7,826 orders might be eligible for shipping refunds or discount vouchers to compensate for the delay

Freight Cost Analysis

```sql
select round(avg(freight_value),2) as avg_freight_cost
from order_items;
```

```
/* avg_freight_cost '19.98' */
```

→ If the average product price is low, a ‘19.98’ freight cost suggests shipping is a significant portion of the total landed cost, potentially impacting customer conversion.

Top 10 Product Category

```sql
select p.product_category_name, round(sum(oi.price),2) as total_sales
from order_items oi
join products p on oi.product_id = p.product_id
group by p.product_category_name
order by total_sales desc
limit 10;
```

```
/* top 10 product categories
 beleza_saude   945719.01
relogios_presentes  933953.78
cama_mesa_banho 802526.27
esporte_lazer   734501.09
informatica_acessorios  695286.77
moveis_decoracao    560031.49
utilidades_domesticas   482852.43
cool_stuff  471536.46
automotivo  449769.11
ferramentas_jardim  376619.95 */
```

→ A large portion of the top 10 is comprised of home-related goods, combined, these categories represent a massive share of consumer interest.

Top 10 Customers by Purchase Value

```
/* Top 10 customers by Purchase value
 0a0a92112bd4c708ca5fde585afaa872   13664.08
da122df9eeddfedc1dc1f5349a1a690c    7571.63
763c8b1c9c68a0229c42c9fc6f662b93    7274.88
dc4802a71eae9be1dd28f5d788ceb526    6929.31
459bef486812aa25204be022145caa62    6922.21
ff4159b92c40ebe40454e3e6a7c35ed6    6726.66
4007669dec559734d6f53e029e360987    6081.54
eebb5dda148d3893cdaf5b5ca3040ccb    4764.34
48e1ac109decbb87765a3eade6854098    4681.78
c8460e4251689ba205045f3ea17884a1    4655.91*/
```

```sql
select c.customer_unique_id, round(sum(op.payment_value),2) as customer_purchase_value
from customers c
join orders o on c.customer_id = o.customer_id
join order_payments op on o.order_id = op.order_id
group by c.customer_unique_id
order by customer_purchase_value desc
limit 10;
```

→The top customer has a significantly higher purchase value compared to the rest of the group. Customers ranked 2nd through 5th show relatively stable spending patterns.